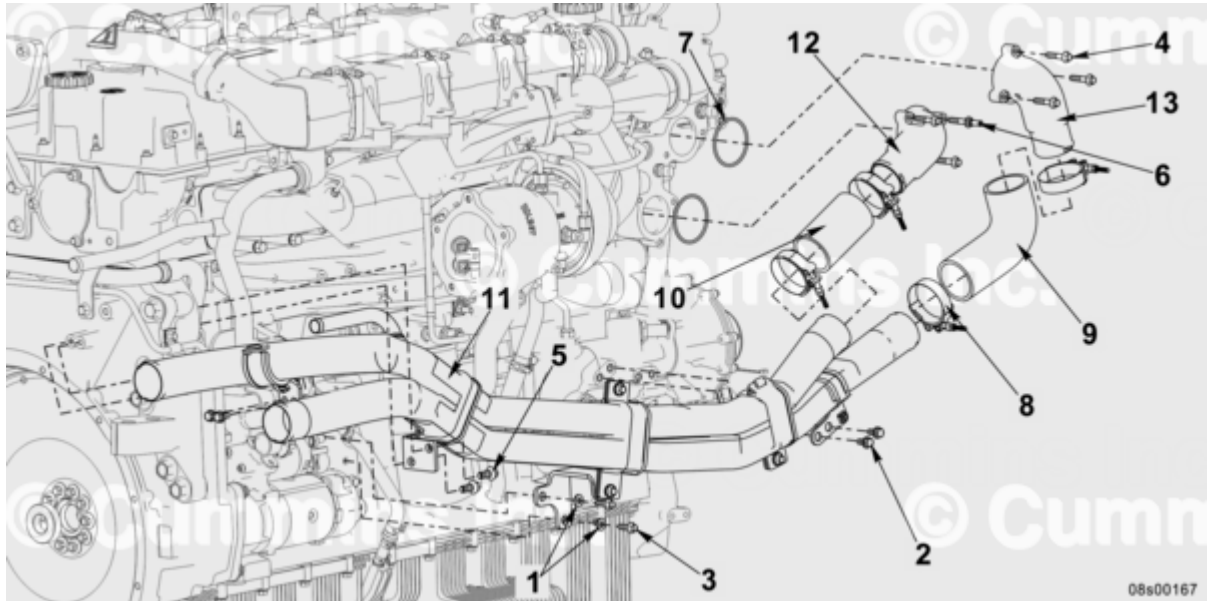


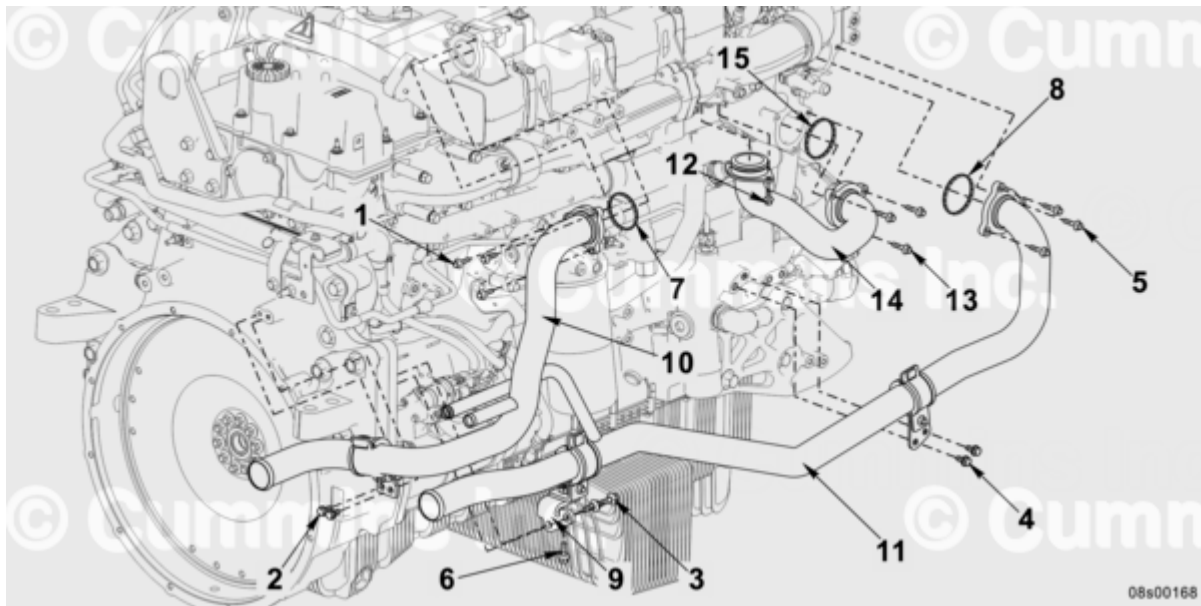
Trainings ▾

Exploded View



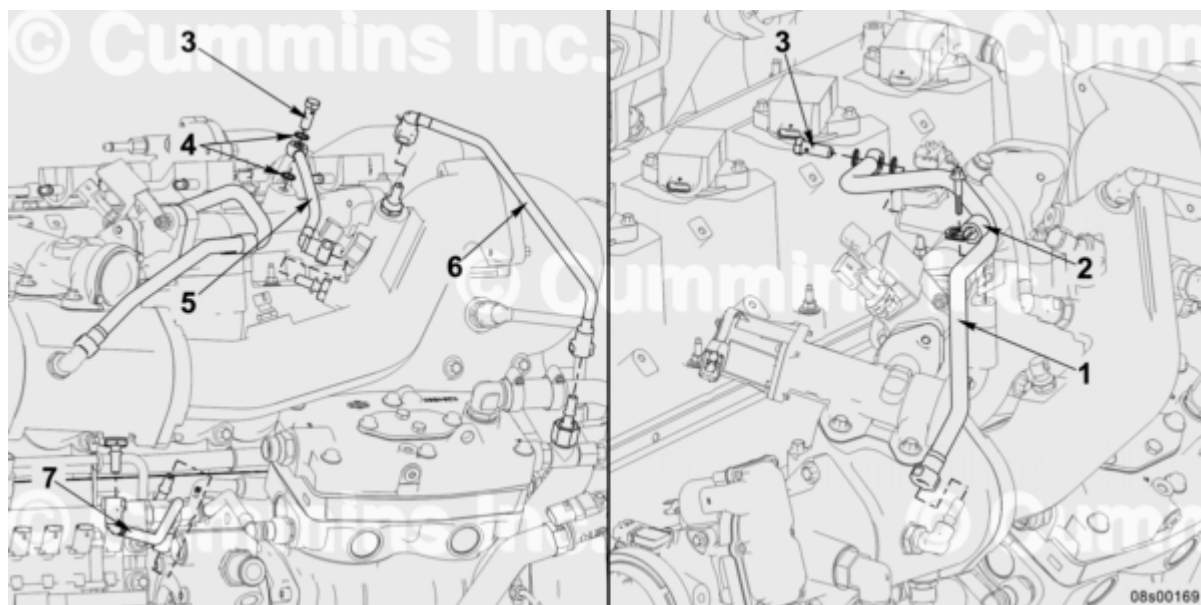
Water Transfer Tube Exploded View (With Retarder, Parallel)

1. Washer - quantity 2
2. Capscrew - quantity 4
3. Capscrew - quantity 2
4. Capscrew - quantity 5
5. Capscrew - quantity 2
6. Stud - quantity 1
7. O-ring - quantity 2
8. Clamp - quantity 4
9. Hose - quantity 1
10. Hose - quantity 1
11. Retarder plumbing - quantity 1
12. Retarder plumbing connection - quantity 1
13. Retarder plumbing connection - quantity 1



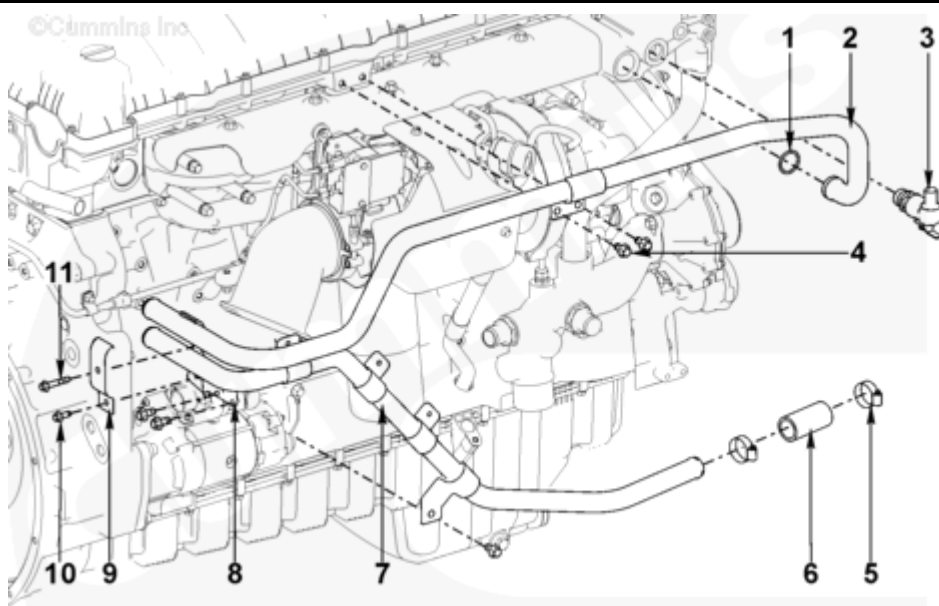
Water Transfer Tube Exploded View (With Retarder, Series)

1. Capscrew - quantity 3
2. Capscrew - quantity 2
3. Capscrew - quantity 2
4. Capscrew - quantity 2
5. Capscrew - quantity 3
6. Capscrew - quantity 1
7. O-ring - quantity 1
8. O-ring - quantity 1
9. Bracket - quantity 1
10. Retarder plumbing - quantity 1
11. Retarder plumbing - quantity 1
12. Capscrew - quantity 1
13. Capscrew - quantity 3
14. Exhaust Gas Recirculation (EGR) cooler inlet tube - quantity 1
15. O-ring - quantity 1



Water Jacket Lines Exploded View

1. Water inlet tube - quantity 1
2. Clip - quantity 1
3. Banjo connector screw - quantity 2
4. O-ring - quantity 4
5. Water outlet tube - quantity 1
6. Water outlet tube - quantity 1
7. Water inlet tube - quantity 1



Water Transfer Tube Exploded View (Without Retarder)

1. O-ring - quantity 1
2. Coolant inlet pipe for urea tank, transmission cooler, and fuel tank heater integration cooling port - quantity 1
3. Cab heater connector - quantity 1
4. Capscrew - quantity 5

5. Clamp - quantity 2
6. Hose - quantity 1
7. Coolant return pipe for urea tank, transmission cooler, and fuel tank heater integration cooling port - quantity 1
8. Bracket - quantity 1
9. Clamp - quantity 1
10. Capscrew - quantity 1
11. Capscrew - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

Additional Service Items

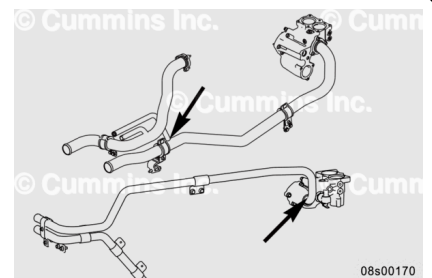
- Lithium-based grease, or equivalent.

General Information

There are two configurations for coolant pipe for urea tank, transmission cooler, and fuel tank heater integration cooling port based on different applications that are with retarder or without retarder.

For the application with retarder, the coolant inlet pipe for urea tank, transmission cooler, and fuel tank heater integration cooling port is connected on the retarder return tube.

For the application without retarder, the coolant inlet pipe for urea tank, transmission cooler, and fuel tank heater integration cooling port is connected on the thermostat housing.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

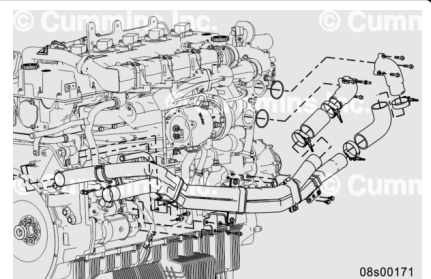
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html)
- Disconnect the fir tree on the coolant return pipe. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html)

Remove

With Retarder, Parallel

Remove the retarder plumbings and the clamps.

Remove the retarder plumbing connections from thermostat housing. Discard the two o-rings.

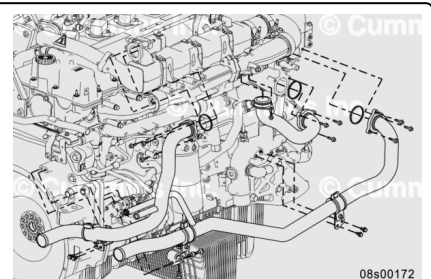


With Retarder, Series

Remove the bracket from flywheel housing.

Remove the retarder plumbings and the clamps. Discard the o-rings.

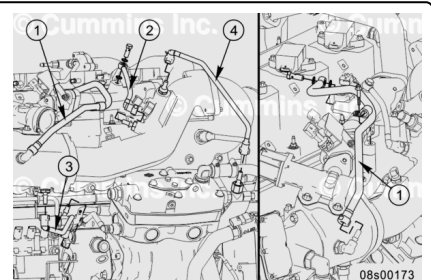
Remove the EGR cooler inlet tube. Discard the two o-rings.



For the water jacket lines, remove the clip. Remove the banjo connector screw. Remove the water inlet tube (1).

Remove the banjo connector screw. Remove the water inlet tube (2).

Remove the water inlet tube (3) and the water outlet tube (4).



Without Retarder

Remove the cab heater connector from the thermostat housing.

Remove the hose and clamps connecting water return pipe to water inlet connection.

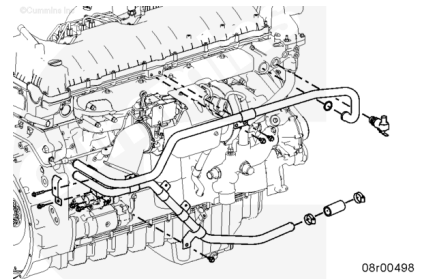
Remove the two capscrews and clamp connecting the ends of two coolant pipes to coolant pipe bracket.

Remove the two capscrews connecting the coolant inlet pipe to cylinder head.

Remove the coolant inlet pipe from thermostat housing.
Discard the o-ring.

Remove the capscrew connecting the coolant return pipe to cylinder block.

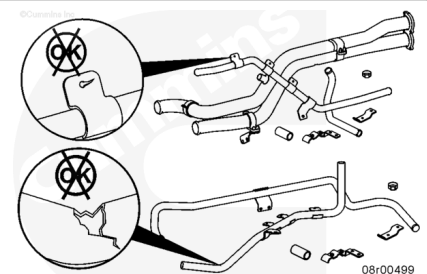
Remove the two capscrews connecting the coolant pipe bracket to flywheel housing.



Inspect

Inspect all hoses, clamp, bracket, and pipes for cracks, cuts, or damage that can cause leaks.

Clean the o-ring groove.



Install

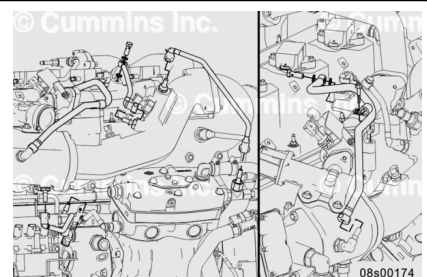
Install the water jacket line connecting venturi tube and air intake connection.

Torque Value: 15 n•m [133 in-lb]

Install the water jacket line connecting venturi tube and EGR mixer.

Torque Value: 15 n•m [133 in-lb]

Install other quick-disconnect tubes.



With Retarder, Parallel

Lubricate two new o-rings with lithium-based grease, or equivalent. Install the retarder plumbing connections to thermostat housing with the two new o-rings. Verify the o-ring is in place and clean.

Torque Value: 18 n•m [159 in-lb]

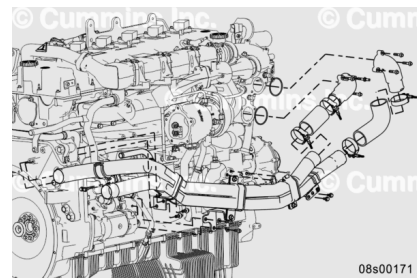
Install the hose and clamps connecting the retarding plumbing connections to the retarder plumbings.

Adjust the retarder flange and tighten four capscrews mounting the retarder plumbings on cylinder block.

Torque Value: 24 n•m [212 in-lb]

Install the four capscrews connecting the retarder plumbings to flywheel housing.

Torque Value: 47 n•m [35 ft-lb]



With Retarder, Series

Lubricate two new o-rings with lithium-based grease, or equivalent. Install the EGR cooler inlet tube to thermostat housing and the EGR cooler with the two new o-rings. Verify the o-ring is in place and clean.

Torque Value: 24 n•m [212 in-lb]

Lubricate a new o-ring with lithium-based grease, or equivalent. Install the retarder plumbing (1) to EGR cooler with the new o-ring. Verify the o-ring is in place and clean.

Torque Value: 24 n•m [212 in-lb]

Install the two capscrews connecting the retarder plumbing (1) to flywheel housing.

Torque Value: 24 n•m [212 in-lb]

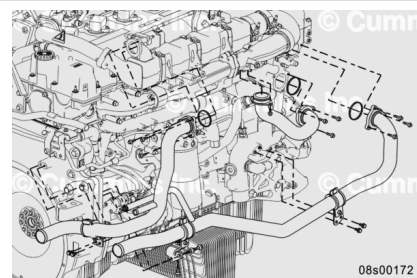
Lubricate a new o-ring with lithium-based grease, or equivalent. Install the retarder plumbing (2) to thermostat housing with the new o-ring. Verify the o-ring is in place and clean.

Torque Value: 18 n•m [159 in-lb]

Install the two capscrews connecting the retarder plumbing (2) to water inlet connection.

Torque Value: 24 n•m [212 in-lb]

Install the bracket on the flywheel housing.



Torque Value: 47 n•m [35 ft-lb]

Install the retarder plumbing (2) on the bracket.

Torque Value: 47 n•m [35 ft-lb]

Without Retarder

Install and tighten the cabin heater connector into the thermostat housing. Adjust the outlet port is directed according to customer requirement.

Torque Value: 68 n•m [50 ft-lb]

The valve **must** be closed.

Install the two capscrews connecting the coolant pipe bracket to flywheel housing.

Torque Value: 45 n•m [33 ft-lb]

Install the coolant return pipe to cylinder block.

Adjust the coolant return pipe flange and tighten two capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the hose and clamps connecting water return pipe to water inlet connection.

Install the coolant inlet pipe to thermostat housing with new o-ring. Verify the o-ring is in place and clean.

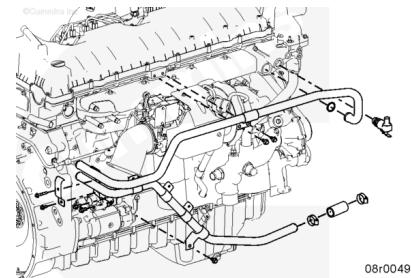
Install the coolant inlet pipe to cylinder head.

Adjust the coolant inlet pipe flange and tighten two capscrews.

Torque Value: 45 n•m [33 ft-lb]

Install the two capscrews and clamp connecting the ends of two coolant pipes to coolant pipe bracket.

Torque Value: 24 n•m [212 in-lb]



Finishing Steps

- Fix the fir tree that is disconnected before to the coolant return pipe. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html>)
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Operate the engine and check for leaks.

Last Modified: 02-Dec-2024
